

SOC L1 ALERT Triage

Here's the link for review of my completed training:

https://tryhackme.com/room/socl1alertrriage?utm_campaign=social_share&utm_medium=social&utm_content=share-completed-room&utm_source=copy&sharerId=646b8859e77c24004a97600b

SOC L1 Alert Triage ⓘ

Learn more about SOC alerts and build a systematic approach to efficiently triaging them.

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1033

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Task 1 ✓ Introduction

An alert is a core concept for any SOC team, and knowing how to handle it properly ultimately decides whether a security breach is detected and prevented, or missed and devastating.

Task 1 ✓ Introduction

An alert is a core concept for any SOC team, and knowing how to handle it properly ultimately decides whether a security breach is detected and prevented, or missed and devastating. This is an entry level but essential room for SOC L1 analysts to understand the concept and lifecycle of alerts, from event generation to correct resolution.

Learning Objectives

- Familiarise with the concept of SOC alert
- Explore alert fields, statuses, and classification
- Learn how to perform alert triage as an L1 analyst
- Practice with real alerts and SOC workflows
- Prepare for SOC Simulator and SAL1 certification

Prerequisites

TryHackMe SIEM

Assigned alert

You haven't picked up any alerts! Assign yourself to an alert, move it to "In Progress", and start the triage!

Search for an alert

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	✎ ↕
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Awaiting action	None	None	✎ ↕
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Awaiting action	None	None	✎ ↕
Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	False Positive	T.Ross (L1)	✎ ↕
Mar 21st 2025 at 11:53	Bruteforce Attack from External	Medium	Closed	True Positive	J.Adams (L2)	✎ ↕

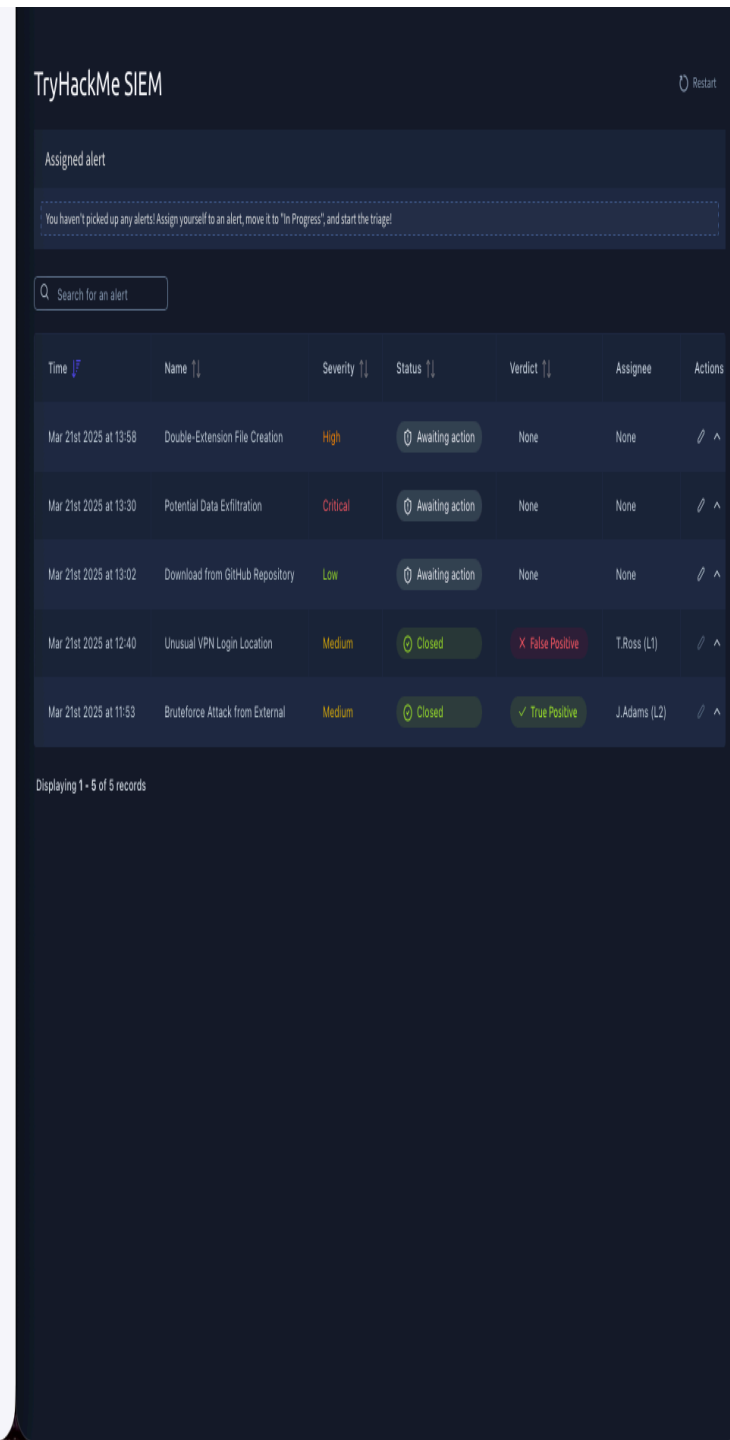
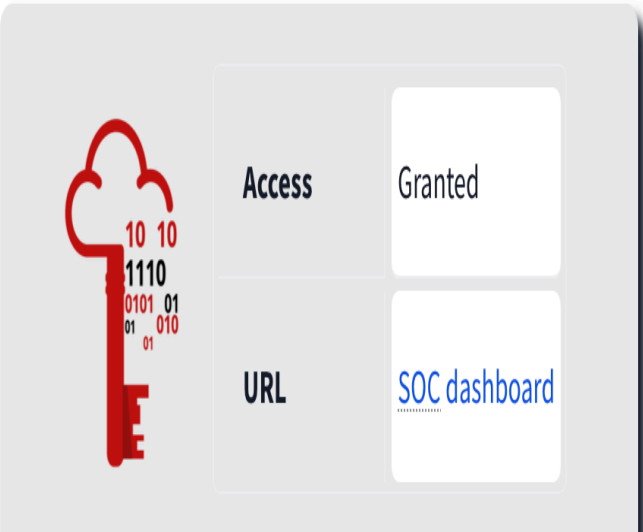
Displaying 1 - 5 of 5 records

Prerequisites

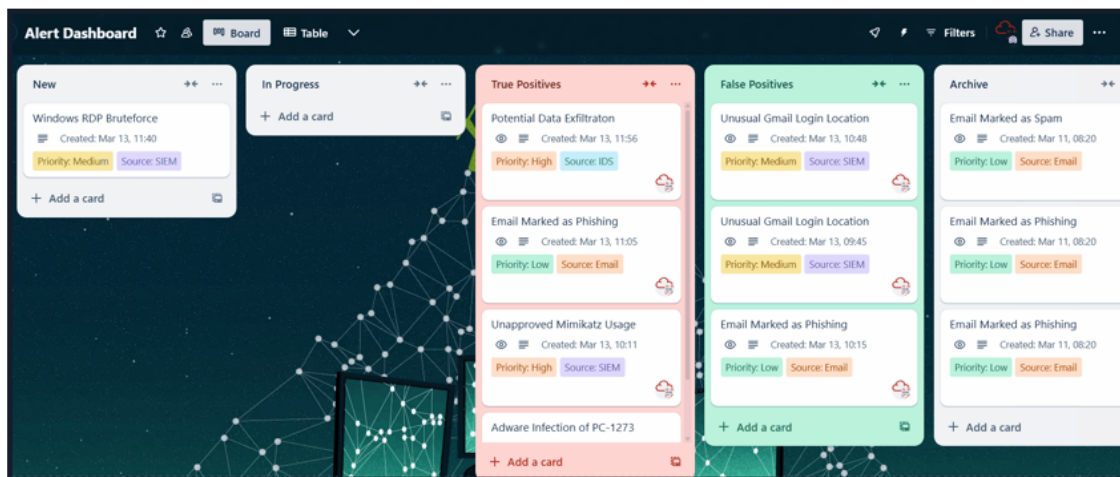
- Understand common attacks on networks, Windows, and Linux
- Know SOC roles and duties, especially of L1 analysts

SOC Dashboard

You were granted access to the SOC dashboard in the TryHackMe SIEM, and you will need it to complete most of the tasks. Open the attached website in a separate window, familiarise yourself with it, and move on to the next task!



Imagine yourself as a SOC intern, looking at the monitor of your mentor, a SOC L1 or L2 analyst, and seeing hundreds of alerts appear, moving between statuses and eventually disappearing from some dashboard or console. You notice lots of alerts named "Email Marked as Phishing", a few "Unusual Gmail Login Location", and one scary "Unapproved Mimikatz Usage" inside a red column. What are these alerts, how are they formed, and what should we do with them? Let's find out!



From Events to Alerts

First, an event must occur, like user login, process launch, or file download. Then, the system, like your OS, a firewall, or a cloud provider must log the event. After that, all system logs must be shipped to a security solution like SIEM or EDR. The SOC team can receive millions of logs per day from thousands of different systems, where most of the events are expected, but some require attention. **Alert**, a notification generated by a security solution when a specific event or sequence of events occurs, is what saves SOC analysts from manual log review by highlighting only suspicious, anomalous events. With alerts, analysts triage just dozens of alerts per day instead of millions of raw logs.

Alert Management Platforms

Solution	Examples	Description
SIEM System	Splunk ES, Elastic	SIEM have solid alert management capabilities and are a perfect choice for most SOC teams
EDR or NDR	MS Defender, CrowdStrike	While EDR and NDR provide their own alert dashboards, it is preferred to use SIEM or SOAR
SOAR System	Splunk SOAR, Cortex SOAR	Bigger SOC teams can use SOAR to aggregate and centralise alerts from multiple solutions

ITSM System	Jira, TheHive	Some teams may have a custom ticket management (ITSM) setup using a dedicated solution (The GIF above is taken from Trello, a simple tool that can be adapted to ITSM needs)
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L1 Role in Alert Triage

SOC L1 analysts are the first line of defence, and they are the ones who work with alerts the most. Depending on various factors, L1 analysts may receive zero to a hundred alerts a day, every one of which can reveal a cyberattack. Still, everyone in the SOC team is somehow involved in the alert triage:

- **SOC L1 analysts:** Review the alerts, distinguish bad from good, and notify L2 analysts in case of a real threat
- **SOC L2 analysts:** Receive the alerts escalated by L1 analysts and perform deeper analysis and remediation
- **SOC engineers:** Ensure the alerts contain enough information required for efficient alert triage
- **SOC manager:** Track speed and quality of alert triage to ensure that real attacks won't be missed

Answer the questions below

What is the number of alerts you see in the [SOC dashboard](#) (opens in new tab)?

5

✓ Correct Answer

5

✓ Correct Answer

What is the name of the most recent alert you see?

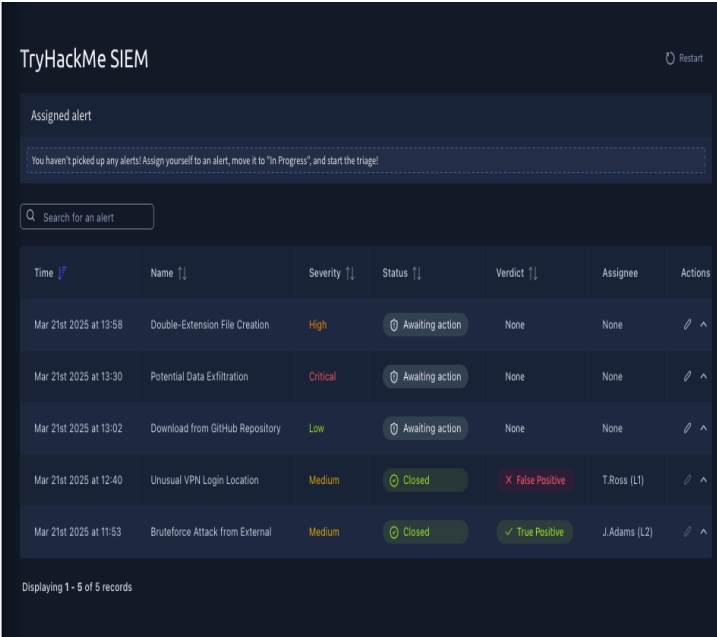
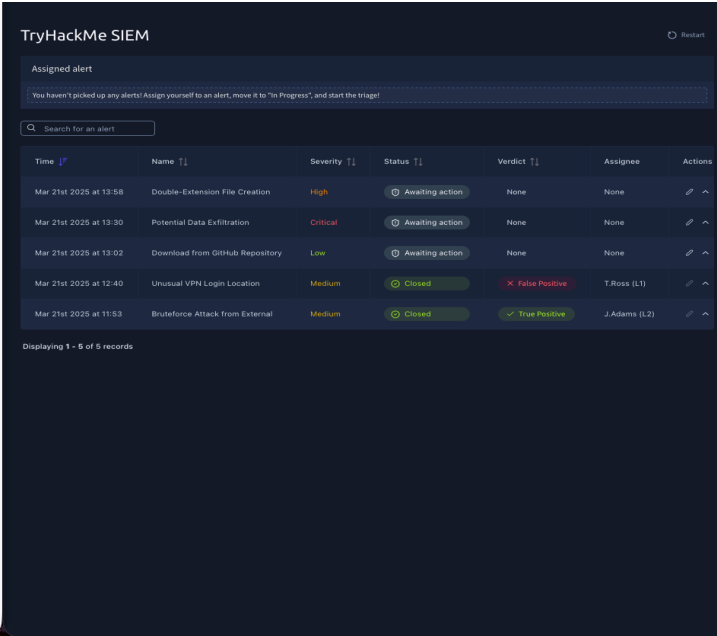
Double-Extension File Creation

✓ Correct Answer

Task 3 ☐ Alert Properties

Task 4 ☐ Alert Prioritisation

Task 5 ☐ Alert Triage



Now that you know how alerts are generated, it's time to review their content. While the details differ for every SIEM or security solution, the alerts generally have a few main properties you must understand before taking them. Don't worry if you find some confusing, as you will hear more about some in the upcoming tasks.

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 11:53	Bruteforce Attack from External	Medium	Closed	True Positive	J.Adams (L2)	
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	
Description:		This rule detects a creation of a double-extension file like *.pdf.exe or *.gif.lnk, often used by hackers in phishing attacks to trick users into opening the malicious executable.				
Host:		LPT-HR-009				
Process Name:		chrome.exe				
Process User:		S.Conway				
Target File:		C:\Users\S.Conway\Downloads\cats2025.mp4.exe				
File MD5:		14d8486f3f63875ef93cf240c5dc10b				

Alert Properties

Nº	Property	Description	Examples
1	Alert Time	Shows alert creation time. Alert usually triggers a few minutes after the actual event	- Alert Time: March 21, 15:35 - Event Time: March 21, 15:32
2	Alert Name	Provides a summary of what happened, based on the detection rule's name	- Unusual Login Location - Email Marked as Phishing - Windows RDP Bruteforce - Potential Data Exfiltration
3	Alert Severity	Defines the urgency of the alert, initially set by detection rule	- Low / Informational - Medium /

TryHackMe SIEM

Assigned alert

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Search for an alert

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	False Positive	T.Ross (L1)	
Description:		The user accessed corporate VPN from a first-seen location. This login may indicate that the user's account is compromised and threat actors are breaking in.				
Source IP:		45.8.112.7				
Source User:		M.Clark				
Login Country:		Japan				
Expected Country:		United States				
Comment:		M.Clark, corporate CFO, confirmed that she is currently on vacation in Japan but had to access VPN from her laptop exactly at that time to resolve an urgent issue. Not a threat				
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Awaiting action	None	None	
Description:		This rule detects 5 or more gigabytes of data sent from a single device to a single destination within a day, which may indicate data exfiltration to untrusted location.				
Destination:		*.zoom.us				
Source IP:		192.168.45.66				
Source Network:		UK04/MEETINGROOM				
Sent Data:		5.8 GB				
Received Data:		5.2 GB				
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Awaiting action	None	None	
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	
Mar 21st 2025 at 11:53	Bruteforce Attack from External	Medium	Closed	True Positive	J.Adams (L2)	

Displaying 1 - 5 of 5 records

3	Alert Severity	Defines the urgency of the alert, initially set by detection engineers, but can be altered by analysts if needed	- Low / Informational - Medium / Moderate - High / Severe - Critical / Urgent
4	Alert Status	Informs if somebody is working on the alert or if the triage is done	- New / Unassigned - In Progress / Pending - Closed / Resolved - And often other custom statuses
5	Alert Verdict	Also called alert classification, explains if the alert is a real threat or noise	- True Positive / Real Threat - False Positive / No Threat - And often other custom verdicts
6	Alert Assignee	Shows the analyst that was assigned or assigned themselves to review the alert	- Assignee can sometimes be called alert owner - Assignee takes responsibility for their alerts
7	Alert Description	Explains what the alert is about, usually in three sections on the right	- The logic of the alert generating rule - Why this activity can indicate an attack - Optionally, how to triage this alert
8	Alert Fields	Provides SOC analysts' comments and values on which the alert was triggered	- Affected Hostname - Entered Commandline - And many more, depending on the alert

Answer the questions below

What was the verdict for the "Unusual VPN Login Location" alert?

False Positive

✓ Correct Answer

What user was mentioned in the "Unusual VPN Login Location" alert?

M.Clark

✓ Correct Answer



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Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	False Positive	T.Ross (L1)	
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Awaiting action	None	None	
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Awaiting action	None	None	

Description: This rule detects any download from GitHub. While GitHub stores lots of great projects that our IT team uses, it also stores malicious scripts and exploits that must not be downloaded by the users.
 Accessed URL: https://github.com/facebook/react
 Source User: G.Chandler
 Source Host: LPT-IT-063
 Source Network: VPN/DEVELOPERS

Mar 21st 2025 at 13:58 Double-Extension File Creation High Awaiting action None None

Description: This rule detects a creation of a double-extension file like ".pdf.exe" or ".gif.link", often used by hackers in phishing attacks to trick users into opening the malicious executable.
 Host: LPT-HR-009
 Process Name: chrome.exe
 Process User: S.Conway
 Target File: C:\Users\S.Conway\Downloads\cats2025.mp4.exe
 File Md5: https://freecatvideohd.monster/cats2025.mp4.exe
 File MD5: 14d8486f3f63875ef93cf6240c5dc10b

Mar 21st 2025 at 11:53 BruteForce Attack from External Medium Closed True Positive J.Adams (L2)

Displaying 1 - 5 of 5 records

TryHackMe SIEM

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Time	Name	Severity	Status	Verdict	Assignee	Actions
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Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Awaiting action	None	None	
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Awaiting action	None	None	
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	
Mar 21st 2025 at 11:53	BruteForce Attack from External	Medium	Closed	True Positive	J.Adams (L2)	

Description: The rule detects 50 or more failed logins on a single system from an external IP during a short period of time. This alert may indicate that a system is Internet-exposed and is actively bruteforced by a threat actor.
 Source IP: 45.148.10.50
 Targeted Users: Administrator,admin,admin
 Host Under Attack: WIN-ITDEV
 BruteForce Method: RDP
 Count of Logins: 1620
 Comment: It was concluded that WIN-ITDEV is a temporary VM created by IT for development purposes, that was not properly secured and exposed RDP port to the Internet, now being bruteforced by a malicious botnet. Although admin password was not guessed, the alert was escalated to L2 to contact IT team

Displaying 1 - 5 of 5 records

TryHackMe SIEM

Assigned alert

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Search for an alert

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	False Positive	T.Ross (L1)	

Description: The user accessed corporate VPN from a first-seen location. This login may indicate that the user's account is compromised and threat actors are breaking in.
 Source IP: 45.8.192.7
 Source User: M.Clark
 Login Country: Japan
 Expected Country: United States
 Comment: M.Clark, corporate CFO, confirmed that she is currently on vacation in Japan but had to access VPN from her laptop exactly at that time to resolve an urgent issue. Not a threat

Mar 21st 2025 at 13:30 Potential Data Exfiltration Critical Awaiting action None None

Mar 21st 2025 at 13:02 Download from GitHub Repository Low Awaiting action None None

Mar 21st 2025 at 13:58 Double-Extension File Creation High Awaiting action None None

Mar 21st 2025 at 11:53 BruteForce Attack from External Medium Closed True Positive J.Adams (L2)

Task 4 ○ Alert Prioritisation



Okay, you can now read and understand the alert. What's next? Recall the second task, where you see hundreds of alerts but have to choose which to pick up first. The process of deciding which to take is called **Alert Prioritisation**, and it is crucial to ensure timely detection of a threat, especially with many alerts in the queue.



Picking the Right Alert

Every SOC team decides on its own prioritisation rules and usually automates them by setting the appropriate alert sorting logic in SIEM or EDR. Below, you may see the generic, simplest, and most commonly used approach:

1. Filter the alerts

Make sure you don't take the alert that other analysts have already reviewed, or that is already being investigated by one of your teammates. You should only take new, yet unseen and unresolved alerts.

2. Sort by severity

Start with critical alerts, then high, medium, and finally low. This is because detection engineers design rules so that critical alerts are much more likely to be real, major threats and cause much more impact than medium or low ones.

3. Sort by time

Start with the oldest alerts and end with the newest ones. The idea is that if both alerts are about two breaches, the hacker from the older breach is likely already dumping your data, while the "newcomer" has just started the discovery.

Answer the questions below

Should you first prioritise medium over low severity alerts? (Yea/Nay)

Yea

✓ Correct Answer

Should you first take the newest alerts and then the older ones? (Yea/Nay)

Nay

✓ Correct Answer

Assign yourself to the first-priority alert and change its status to **In Progress**.

The name of your selected alert will be the answer to the question.

TryHackMe SIEM

Assigned alert

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Search for an alert

Time	Name	Severity	Status	Verdict	Assignee	Actions
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Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	False Positive	T.Ross (L1)	
Description: The user accessed corporate VPN from a first-seen location. This login may indicate that the user's account is compromised and threat actors are breaking in. Source IP: 45.8.112.7 Source User: M.Clark Login Country: Japan Expected Country: United States Comment: M.Clark, corporate CFO, confirmed that she is currently on vacation in Japan but had to access VPN from her laptop exactly at that time to resolve an urgent issue. Not a threat						
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Awaiting action	None	None	
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Awaiting action	None	None	
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	

Displaying 1 - 5 of 5 records

TryHackMe SIEM

Assigned alert

You haven't picked up any alerts! Assign yourself to an alert, move it to "In Progress", and start the triage!

Search for an alert

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 11:53	Bruteforce Attack from External	Medium	Closed	True Positive	J.Adams (L2)	
Mar 21st 2025 at 12:40	Unusu				T.Ross (L1)	
Description: The user accessed corporate VPN from a first-seen location. This login may indicate that the user's account is compromised and threat actors are breaking in. Source IP: 45.8.112.7 Source User: M.Clark Login Country: Japan Expected Country: United States Comment: M.Clark, corporate CFO, confirmed that she is currently on vacation in Japan but had to access VPN from her laptop exactly at that time to resolve an urgent issue. Not a threat						
Mar 21st 2025 at 13:02	Downl				None	
Mar 21st 2025 at 13:30	Potent				None	
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	

Displaying 1 - 5 of 5 records

✓ Correct Answer

Assign yourself to the first-priority alert and change its status to **In Progress**.

The name of your selected alert will be the answer to the question.

Potential Data Exfiltration

✓ Correct Answer



Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Awaiting action	None	None		
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	In Progress	None	You (L1)		
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None		

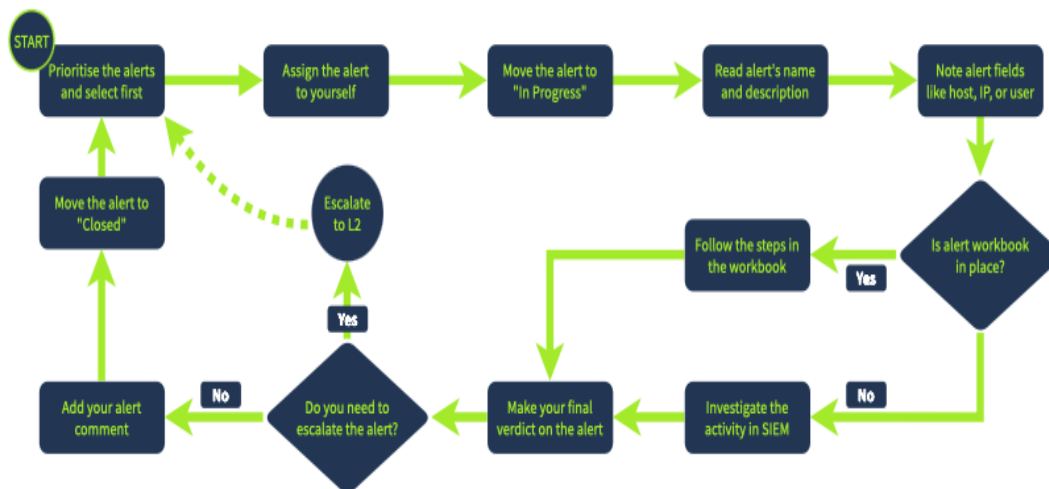
Displaying 1 - 5 of 5 records

Task 5 ○ Alert Triage



Finally, you are ready to review the chosen alert! The process is quite operationally heavy, but you will soon see why every step is important. Also, note that the alert review by SOC analysts can also be called alert triage, alert handling, alert processing, alert investigation, or alert analysis. During this module, we will stick to the **Alert Triage** option.

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A reference flow for alert triage in TryHackMe labs.

Note that the flow may vary slightly depending on the team.

Initial Actions

The initial steps are designed to ensure that you take ownership of the assigned alert and avoid interfering with alerts being handled by other analysts, and confirm that you are fully prepared to proceed with the detailed investigation. You achieve it by first assigning the alert to yourself, moving it to **In Progress**, and then familiarising yourself with the alert details like its name, description, and key indicators.

Investigation

This is the most complex step, requiring you to apply your technical knowledge and experience to understand the activity and properly analyse its legitimacy in SIEM or EDR logs. To support L1 analysts with this step, some teams develop **Workbooks** (also known as playbooks or runbooks) - instructions on how to investigate the specific category of alerts. If workbooks are not available, below are some key recommendations:

1. Understand who is under threat, like the affected user, hostname, cloud, network, or website
2. Note the action described in the alert, like whether it was a suspicious login, malware, or phishing
3. Review surrounding events, looking for suspicious actions shortly after or before the alert
4. Use threat intelligence platforms or other available resources to verify your thoughts

Final Actions

Your decisions here determine whether you found or missed the potential cyberattack. Some actions like **Escalation** or **Commenting** will be explained in the following rooms, so don't worry if they sound complex right now. First, decide if the alert you investigated is malicious (True Positive) or not (False Positive). Then, prepare your detailed comment explaining your analysis steps and verdict reasoning, return to the dashboard and move it to the **Closed** status.

SOC Dashboard Notes

If you didn't receive a flag after your triage, it means that the values you set are wrong.

You can reset the SOC dashboard by clicking **Restart** on the top right in the TryHackMe SIEM.

TryHackMe SIEM

Restart

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 11:53	Bruteforce Attack from External	Medium	Closed	True Positive	J.Adams (L2)	
Description: The rule detects 50 or more failed logins on a single system from an external IP during a short period of time. This alert may indicate that a system is Internet-exposed and is actively bruteforced by a threat actor. Source IP: 45.148.10.50 Targeted Users: Administrator,admin,adm Host Under Attack: WIN-ITDEV Bruteforce Method: RDP Count of Logins: 1620 Comment: It was concluded that WIN-ITDEV is a temporary VM created by IT for development puposes, that was not properly secured and exposed RDP port to the Internet, now being bruteforced by a malicious botnet. Although						
Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	False Positive	T.Ross (L1)	
Description: This rule detects any download from GitHub. While GitHub stores lots of great projects that our IT team uses, it also stores malicious scripts and exploits that must not be downloaded by the users. Accessed URL: https://github.com/facebook/react Source User: G.Chandler Source Host: LPT-IT-063 Source Network: VPN/DEVELOPERS Comment: It was determined that it is a false positive due to G. Chandler downloading a file from Github.						
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Closed	False Positive	You (L1)	
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Closed	True Positive	You (L1)	
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	

Displaying 1 - 5 of 5 records

Answer the questions below

Which flag did you receive after you correctly triaged the first-priority alert?

THM{looks_like_lots_of_zoom_meetings}

✓ Correct Answer



TryHackMe SIEM

Restart

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 11:53	Bruteforce Attack from External	Medium	Closed	True Positive	J.Adams (L2)	
Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	False Positive	T.Ross (L1)	
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Awaiting action	None	None	
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Closed	False Positive	You (L1)	
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Awaiting action	None	None	

Displaying 1 - 5 of 5 records

Which flag did you receive after you correctly triaged the second-priority alert?

THM{how_could_this_user_fall_for_it?}

✓ Correct Answer

Mar 21st 2025 at 13:58 Double-Extension File Creation High Closed ✓ True Positive You (L1)

Description: This rule detects a creation of a double extension file like 'file.exe.tmp', often used by hackers in

Host: THM{how_could_this_user_fall_for_it?}

Process Name: C:\Users\S Conway\Downloads\cats2025.mp4.exe

Process User: https://freecatvideoshd.monster/cats2025.mp4.exe

Target File: 14d848613f63875ef93cd4240c5dc10b

File MD5: It is determined that that this alert is a true positive due to cats2025.exe executive file downloaded by S. Conway from a phishing attack by the malicious link the unknown user LPT-HR-009 sent.

Comment: Displaying 1 - 5 of 5 records

Which flag did you receive after you correctly triaged the third-priority alert?

THM{should_we_allow_github_for_dev}

✓ Correct Answer

TryHackMe SIEM Restart

Search for an alert

Time	Name	Severity	Status	Verdict	Assignee	Actions
Mar 21st 2025 at 11:53	BruteForce Attack from External	Medium	Closed	✓ True Positive	J.Adams (L2)	🔍 ⬆ ⬇
Mar 21st 2025 at 12:40	Unusual VPN Login Location	Medium	Closed	✗ False Positive	T.Ross (L1)	🔍 ⬆ ⬇
Mar 21st 2025 at 13:02	Download from GitHub Repository	Low	Closed	✗ False Positive	You (L1)	🔍 ⬆ ⬇
Mar 21st 2025 at 13:30	Potential Data Exfiltration	Critical	Closed	✗ False Positive	You (L1)	🔍 ⬆ ⬇
Mar 21st 2025 at 13:58	Double-Extension File Creation	High	Closed	✓ True Positive	You (L1)	🔍 ⬆ ⬇

Displaying 1 - 5 of 5 records

Great work and correct verdict! Claim your flag!

THM{should_we_allow_github_for_devs?}

Room completed: 100%

Task 6 ✓ Conclusion

Congratulations on successfully triaging the alerts! Of course, closing the alert as True Positive won't prevent the attack, but it is a great start. Next, you will learn about proper alert commenting and case reporting, correct escalation procedures, and actions made by L2 analysts after the escalation. We hope you enjoyed the room!

Answer the questions below

I am ready to move on!

No answer needed

✓ Correct Answer